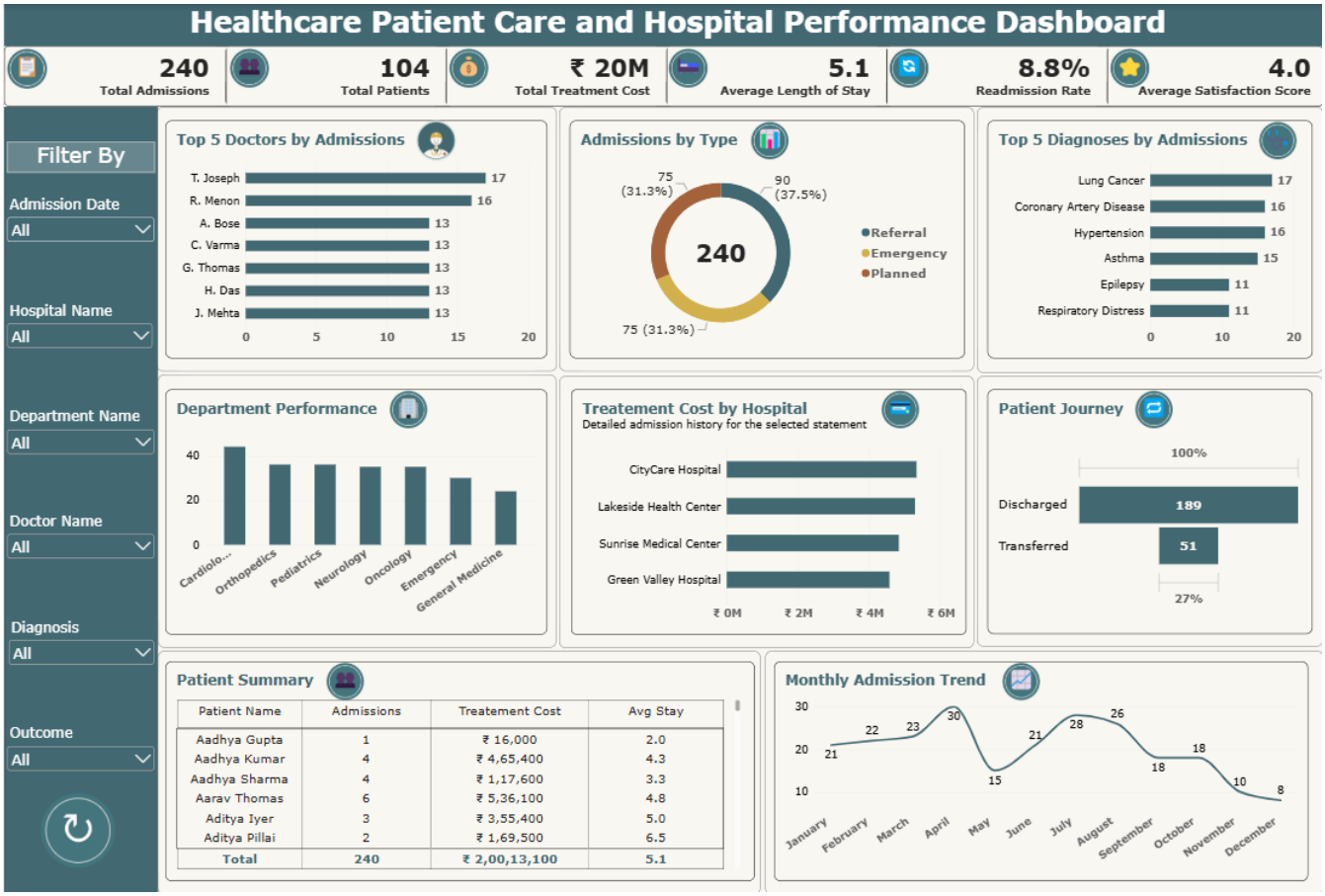
Healthcare Patient Care and Hospital Performance Dashboard

Power BI Project Report



Patient Name	Admissions	Treatement Cost	Avg Stay
Aadhya Gupta	1	₹ 16,000	2.0
Aadhya Kumar	4	₹ 4,65,400	4.3
Aadhya Sharma	4	₹ 1,17,600	3.3
Aarav Thomas	6	₹ 5,36,100	4.8
Aditya Iyer	3	₹ 3,55,400	5.0
Aditya Pillai	2	₹ 1,69,500	6.5
Total	240	₹ 2,00,13,100	5.1

Hospital Overview dashboard

This report documents a Power BI dashboard for healthcare operations, patient analysis, Drillthrough, bookmarks, tooltips, and Dynamic Row Level Security.

All names, hospitals, emails, and healthcare records are fictional and used for educational purposes.

Project Overview

The dashboard gives hospital users a single interactive view of admissions, patient volume, treatment costs, length of stay, readmission rate, satisfaction, department activity, doctor workload, diagnoses, and patient outcomes. It supports operational monitoring while restricting access to sensitive records based on the user's role.

Project Objectives

- Monitor hospital and department performance through key operational indicators.
- Analyze admissions, treatment costs, diagnoses, doctors, and patient outcomes.
- Enable patient-level investigation without losing the dashboard context.
- Apply role-based access control through Dynamic Row Level Security.

Data Model

Admissions is the central fact table. It is connected to the Hospitals, Departments, Doctors, and Patients dimension tables. UsersSecurity stores user emails, roles, and assigned access identifiers for Dynamic RLS.

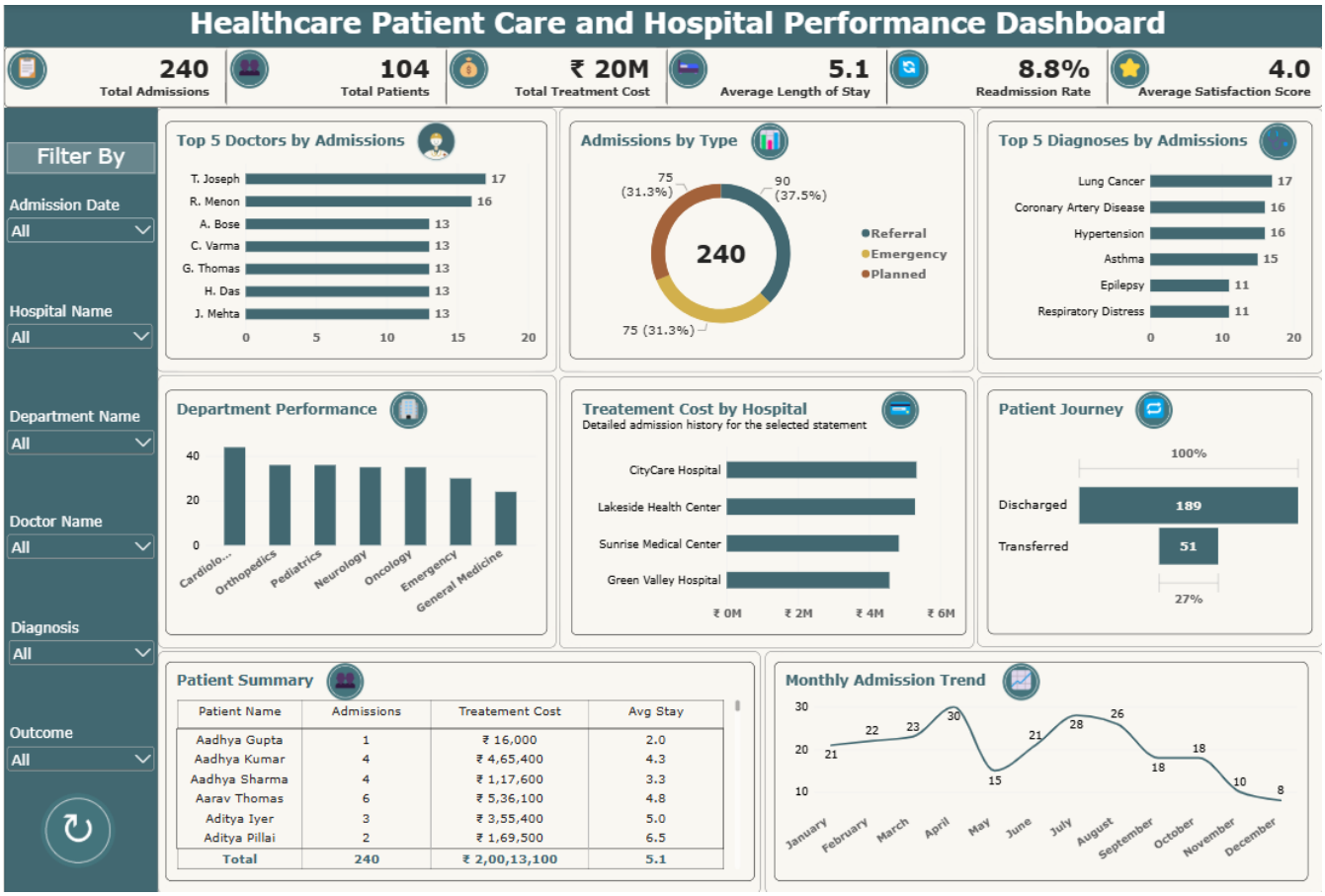
Table	Purpose
Admissions	Fact table containing admission events, dates, costs, outcomes, diagnoses, and related IDs.
Hospitals, Departments, Doctors, Patients	Dimension tables used to filter and group admission results.
UsersSecurity	Security table containing UserEmail, Role, HospitalID, DepartmentID, and DoctorID.

Core Measures

- Total Admissions
- Total Patients
- Total Treatment Cost
- Average Length of Stay
- Readmission Rate
- Average Satisfaction Score

Hospital Overview Dashboard

The Hospital Overview page is designed for management-level monitoring. The left-side slicers allow users to filter results by admission date, hospital, department, doctor, diagnosis, and outcome.



Patient Name	Admissions	Treatement Cost	Avg Stay
Aadhya Gupta	1	₹ 16,000	2.0
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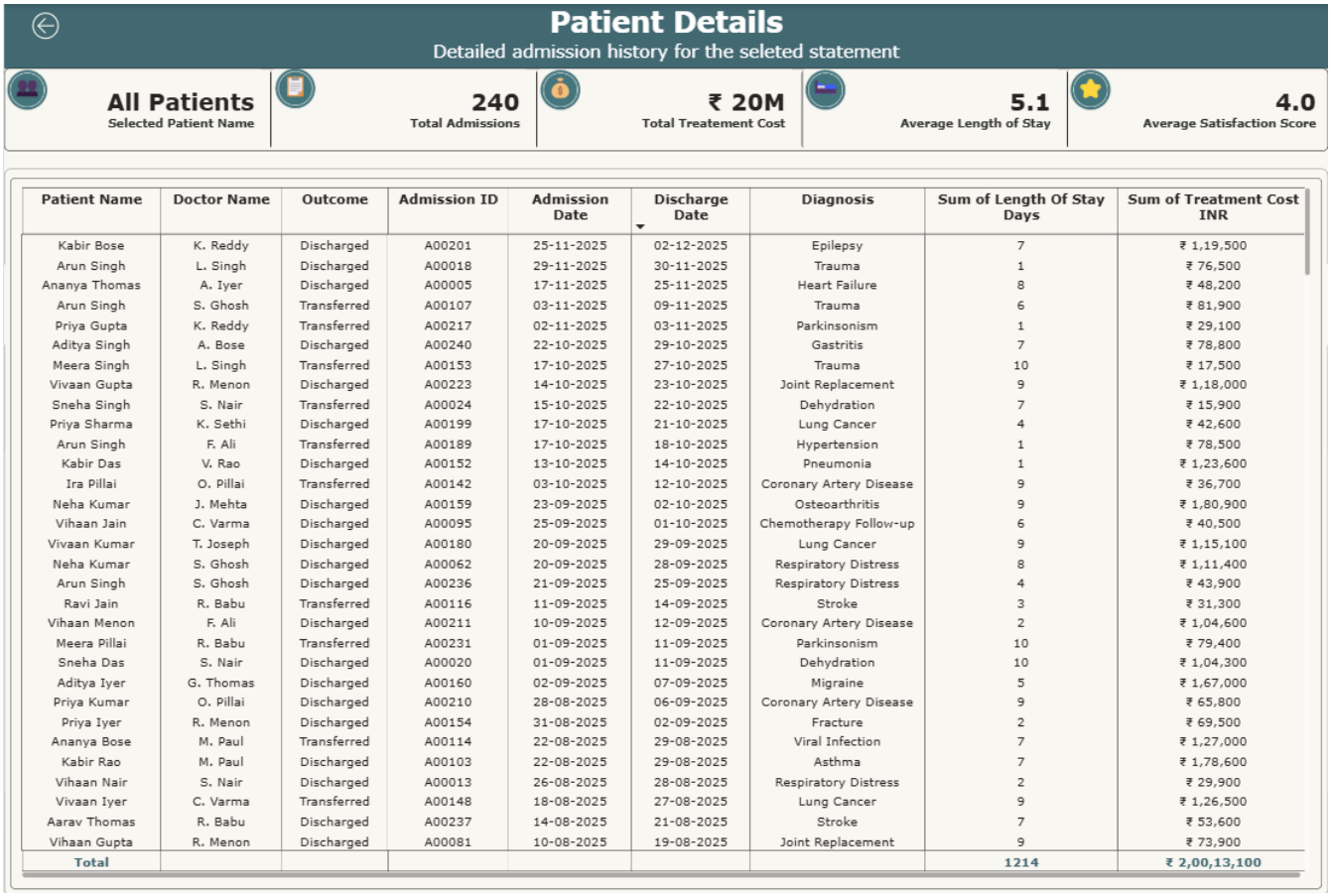
Figure 1. Hospital Overview with KPIs, filters, analysis visuals, and Patient Summary

Visual Analysis

Visual	Business Purpose
Top 5 Doctors by Admissions	Highlights doctors with the largest admission volume.
Admissions by Type	Compares emergency, planned, and referral admissions.
Top 5 Diagnoses	Identifies the most common diagnoses in the selected context.
Department Performance	Compares admission volume across departments.
Treatment Cost by Hospital	Shows relative treatment costs across hospitals.
Patient Journey	Compares discharged and transferred patient outcomes.
Monthly Admission Trend	Tracks changes in admissions over time.

Patient Details Drillthrough

The Patient Details page provides detailed admission history for the selected patient. Users reach this page by right-clicking a patient in the Patient Summary table and choosing Drillthrough. The page retains the selected context and provides a Back button to return to Hospital Overview.



Patient Name	Doctor Name	Outcome	Admission ID	Admission Date	Discharge Date	Diagnosis	Sum of Length Of Stay Days	Sum of Treatment Cost INR
Kabir Bose	K. Reddy	Discharged	A00201	25-11-2025	02-12-2025	Epilepsy	7	₹ 1,19,500
Arun Singh	L. Singh	Discharged	A00018	29-11-2025	30-11-2025	Trauma	1	₹ 76,500
Ananya Thomas	A. Iyer	Discharged	A00005	17-11-2025	25-11-2025	Heart Failure	8	₹ 48,200
Arun Singh	S. Ghosh	Transferred	A00107	03-11-2025	09-11-2025	Trauma	6	₹ 81,900
Priya Gupta	K. Reddy	Transferred	A00217	02-11-2025	03-11-2025	Parkinsonism	1	₹ 29,100
Aditya Singh	A. Bose	Discharged	A00240	22-10-2025	29-10-2025	Gastritis	7	₹ 78,800
Meera Singh	L. Singh	Transferred	A00153	17-10-2025	27-10-2025	Trauma	10	₹ 17,500
Vivaan Gupta	R. Menon	Discharged	A00223	14-10-2025	23-10-2025	Joint Replacement	9	₹ 1,18,000
Sneha Singh	S. Nair	Transferred	A00024	15-10-2025	22-10-2025	Dehydration	7	₹ 15,900
Priya Sharma	K. Sethi	Discharged	A00199	17-10-2025	21-10-2025	Lung Cancer	4	₹ 42,600
Arun Singh	F. Ali	Transferred	A00189	17-10-2025	18-10-2025	Hypertension	1	₹ 78,500
Kabir Das	V. Rao	Discharged	A00152	13-10-2025	14-10-2025	Pneumonia	1	₹ 1,23,600
Ira Pillai	O. Pillai	Transferred	A00142	03-10-2025	12-10-2025	Coronary Artery Disease	9	₹ 36,700
Neha Kumar	J. Mehta	Discharged	A00159	23-09-2025	02-10-2025	Osteoarthritis	9	₹ 1,80,900
Vihaan Jain	C. Varma	Discharged	A00095	25-09-2025	01-10-2025	Chemotherapy Follow-up	6	₹ 40,500
Vivaan Kumar	T. Joseph	Discharged	A00180	20-09-2025	29-09-2025	Lung Cancer	9	₹ 1,15,100
Neha Kumar	S. Ghosh	Discharged	A00062	20-09-2025	28-09-2025	Respiratory Distress	8	₹ 1,11,400
Arun Singh	S. Ghosh	Discharged	A00236	21-09-2025	25-09-2025	Respiratory Distress	4	₹ 43,900
Ravi Jain	R. Babu	Transferred	A00116	11-09-2025	14-09-2025	Stroke	3	₹ 31,300
Vihaan Menon	F. Ali	Discharged	A00211	10-09-2025	12-09-2025	Coronary Artery Disease	2	₹ 1,04,600
Meera Pillai	R. Babu	Transferred	A00231	01-09-2025	11-09-2025	Parkinsonism	10	₹ 79,400
Sneha Das	S. Nair	Discharged	A00020	01-09-2025	11-09-2025	Dehydration	10	₹ 1,04,300
Aditya Iyer	G. Thomas	Discharged	A00160	02-09-2025	07-09-2025	Migraine	5	₹ 1,67,000
Priya Kumar	O. Pillai	Discharged	A00210	28-08-2025	06-09-2025	Coronary Artery Disease	9	₹ 65,800
Priya Iyer	R. Menon	Discharged	A00154	31-08-2025	02-09-2025	Fracture	2	₹ 69,500
Ananya Bose	M. Paul	Transferred	A00114	22-08-2025	29-08-2025	Viral Infection	7	₹ 1,27,000
Kabir Rao	M. Paul	Discharged	A00103	22-08-2025	29-08-2025	Asthma	7	₹ 1,78,600
Vihaan Nair	S. Nair	Discharged	A00013	26-08-2025	28-08-2025	Respiratory Distress	2	₹ 29,900
Vivaan Iyer	C. Varma	Transferred	A00148	18-08-2025	27-08-2025	Lung Cancer	9	₹ 1,26,500
Aarav Thomas	R. Babu	Discharged	A00237	14-08-2025	21-08-2025	Stroke	7	₹ 53,600
Vihaan Gupta	R. Menon	Discharged	A00081	10-08-2025	19-08-2025	Joint Replacement	9	₹ 73,900
Total							1214	₹ 2,00,13,100

Figure 2. Patient Details Drillthrough page

Patient Details Content

- Selected patient name and patient-level KPI cards.
- Admission ID, admission date, discharge date, diagnosis, doctor, and outcome.
- Length of stay and treatment cost for each admission record.

Dynamic Row Level Security

Dynamic RLS uses the signed-in Power BI user's email address to determine the records that user can view. The UsersSecurity table maps each UserEmail to a role and assigned HospitalID, DepartmentID, or DoctorID.

Role	Allowed Data
Hospital Admin	All admission records.
Hospital Manager	Admissions from the assigned hospital.
Department Head	Admissions from the assigned department.
Doctor	Admissions assigned to that doctor.

Example Dynamic RLS Logic

```
Admissions[HospitalID] =
LOOKUPVALUE(
    UsersSecurity[HospitalID],
    UsersSecurity[UserEmail],
    USERPRINCIPALNAME()
)
```

The logic retrieves the logged-in user's assigned HospitalID from UsersSecurity and filters the Admissions table accordingly. Equivalent expressions can be applied for department and doctor-level access.

Bookmarks and Tooltips

The dashboard includes a Reset Filters button connected to a bookmark that restores the default slicer state. Tooltips provide additional context to users without adding unnecessary text to the dashboard canvas.

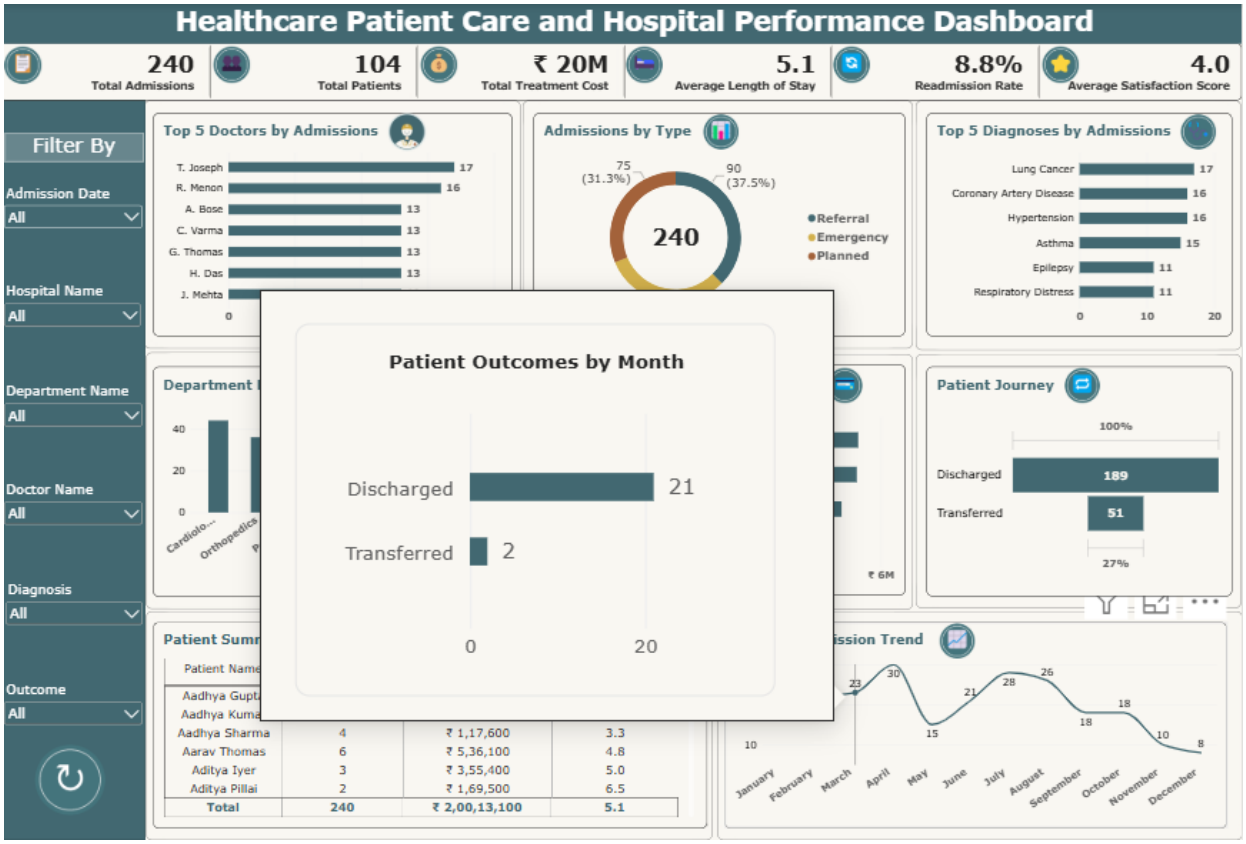


Figure 3. Tooltip example for patient outcomes

Project Benefits

- Brings key healthcare performance indicators into one interactive report.
- Makes trends, costs, department activity, and patient outcomes easier to compare.
- Supports detailed patient investigation through Drillthrough.
- Protects sensitive data by limiting views according to each user role.
- Improves usability with filters, bookmarks, tooltips, and consistent visual design.

Conclusion

This project demonstrates a practical Power BI healthcare analytics solution that combines operational reporting with secure access control. Hospital Overview supports broad performance monitoring, while Patient Details supports record-level investigation. Dynamic RLS ensures users can view only the data relevant to their responsibilities.